SSC MTS 2024 question paper with answers

12 November 2024 · Shift 1

75 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 15 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 13 questions

Q.2

The average score of 12 girls in a test was 16.5 and that of 18 boys in the same test was 16. The average score of girls and boys taken together in the same test was:

(A) 16.7
(B) 16.4
(C) 16.1
(D) 16.2

Q.3

I walk at a speed of 10 km/h and reach the destination in 2 hours. If I increase my speed by 5 km/h, how early would I reach my destination?

(A) 40 min
(B) 20 min
(C) 30 min
(D) 50 min

Q.4

In an office having 160 staff members, 40% are females and the remaining are males. The average weight of females is 58 kg and that of males is 63 kg. Find the average weight of all staff members.

(A) 61 kg
(B) 58 kg
(C) 68 kg
(D) 79 kg

Q.5

Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 4% per year rate of interest for 6 years.

(A) 540
(B) 640
(C) 560
(D) 600

Q.7

A man buys 15 identical articles for a total of ₹15. If he sells each of them for ₹1.32, then his profit percentage is:

(A) 23%
(B) 50%
(C) 8%
(D) 32%

Q.8

A person lent a sum of ₹30,000 on simple interest for 12 years, with an interest rate of 8% per annum for the first 4 years, 10% per annum for the next 3 years, and 15% per annum for last 5 years. How much interest (in ₹) will he earn at the end of 12 years?

(A) 41100
(B) 40200
(C) 40800
(D) 38000

Q.10

The average age of a man and his son is 48 years. The ratio of their ages is 5 : 3. What is the son's age?

(A) 38 years
(B) 35 years
(C) 34 years
(D) 36 years

Q.11

When a number is first increased by 18% and then decreased by 18%, the change in the number is _____.

(A) 0%
(B) 3.24% increase
(C) 3.24 % decrease
(D) 32.4% increase

Q.13

A vehicle moves at 65 km/h for 4 hours before slowing down to 52 km/h for the next 2.5 hours. Calculate the average speed.

(A) 58.5 km/h
(B) 60 km/h
(C) 55 km/h
(D) 61.25 km/h

Q.15

What is the radius of the largest right circular cone that can be cut out from a cube of edge 8.4 cm ?

(A) 2.1 cm
(B) 4.1 cm
(C) 2.2 cm
(D) 4.2 cm

Q.18

A printer with a marked price of ₹9,000 is offered on successive discounts of 30%, 20%, and 15%. Identify the printer's selling price.

(A) ₹4,284
(B) ₹4,250
(C) ₹4,264
(D) ₹4,274

Q.19

The average of seven consecutive natural numbers is 56. The greatest number among them is:

(A) 60
(B) 58
(C) 61
(D) 59

Q.20

What sum (in ₹) will earn a simple interest of ₹240 in 4 years at 20% per year rate of interest?

(A) 600
(B) 150
(C) 450
(D) 300

Reasoning Ability and Problem Solving 14 questions

Q.1

Which two numbers should be interchanged to make the given equation correct? (12÷ 3) × 18 - 2× 4 + 42÷ 6 = 69 (Note: Interchange should be done of entire number and not individual digits of a given number)

(A) 18 and 12
(B) 3 and 4
(C) 3 and 6
(D) 2 and 6

Q.4

Read the given statement(s) and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statement(s). Statements: No lizard is a snake. All lizards are reptiles. Conclusions: (I) At least some reptiles are snakes. (II) All reptiles are lizards.

(A) Only conclusion (I) follows.
(B) Only conclusion (II) follows.
(C) Both conclusions (I) and (II) follow.
(D) Neither conclusion (I) nor (II) follows.

Q.5

In a certain code language, 'you are a good friend' is coded as 'es an pz fd mq' and 'his friend is very kind' is coded as 'ck mq pn ag sb'. How is 'friend' coded in the given language?

(A) mq
(B) fd
(C) ck
(D) es

Q.6

13 is related to 103 following a certain logic. Following the same logic, 9 is related to 71. To which of the following is 15 related, following the same logic? (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding / subtracting / multiplying etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 109
(B) 117
(C) 119
(D) 112

Q.7

JQMT is related to PWSZ in a certain way based on the English alphabetical order. In the same way, HOKR is related to NUQX. To which of the following is FMIP related, following the same logic?

(A) OSLV
(B) LSOV
(C) LOSV
(D) OLSV

Q.9

Seven people A, B, C, D, E, F and G are sitting around a circular table, facing the centre (but not necessarily in the same order). B is sitting second to the left of C. Only two people are sitting between B and D. D is an immediate neighbour of C. A is sitting third to the right of F. E is sitting to the immediate left of G. What is the position of E with respect to B?

(A) Second to the left
(B) Third to the right
(C) To theimmediate left
(D) To the immediate right

Q.10

What should come in place of the question mark (?) in the given series based on the English alphabetical order? AEH, DHK, GKN, JNQ, ?

(A) MQX
(B) PQT
(C) MQT
(D) NRU

Q.11

In a certain code language, 'MOTH' is coded as '2514' and 'TOME' is coded as '1453'. What is the code for 'E' in that language?

(A) 3
(B) 5
(C) 4
(D) 1

Q.12

What should come in place of the question mark (?) in the following series? 3, 7, 15, 31, 63, ?

(A) 168
(B) 576
(C) 177
(D) 127

Q.14

In a code language, 'ROASTER' is coded as '96' and 'GUM' is coded as '41'. How will 'ARTIST' be coded in the same language?

(A) 82
(B) 77
(C) 88
(D) 87

Q.15

What will come in the place of the question mark (?) in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 29 × 841 ÷ 87 + 37 – 59 = ?

(A) 52
(B) 65
(C) 25
(D) 55

Q.16

The second number in the given number pairs is obtained by performing certain mathematical operation(s) on the first number. Select the pair in which the numbers are related in the same way as are the numbers of the given pairs. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding /deleting /multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) 11, 88; 14, 112

(A) 19, 151
(B) 16, 128
(C) 24, 196
(D) 28, 226

Q.18

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: No soap is a lotion. No lotion is a conditioner. Conclusions: (I) No soap is a conditioner. (II) Some conditioners are lotions.

(A) Neither conclusion I nor II follows
(B) Both conclusion I and II follow
(C) Only conclusion II follows
(D) Only conclusion I follows

Q.19

Six persons – L, M, N, O, P and Q – are sitting around a circular table facing the centre, but not necessarily in the same order. O is sitting second to the left of Q. P is not an immediate neighbour of Q. M is sitting second to the right of L. N is sitting to the immediate right of Q. Who is an immediate neighbour of both, L and N?

(A) P
(B) Q
(C) O
(D) M

General Awareness 25 questions

Q.1

Which of the following vitamins includes cobalt as an essential component?

(A) Vitamin A
(B) Vitamin D
(C) Vitamin E
(D) Vitamin B12

Q.2

The construction of Mehrangarh fort of Jodhpur was initiated during the reign of which ruler?

(A) Rao Bika
(B) Rao Jodha
(C) Sawai Man Singh
(D) Maharawal Jaswant Singh

Q.3

_____ was awarded, Padma Shri in 1970 for his contribution to the field of Kuchipudi dance.

(A) Dwaram Venkataswami Naidu
(B) Mushtaq Husain Khan
(C) V Satyanarayana Sarma
(D) Sthanam NarasimhaRao

Q.4

The 11th Fundamental Duty to the Indian Constitution was added in the year _____ by the 86th Constitutional Amendment Act.

(A) 2002
(B) 2004
(C) 2000
(D) 2001

Q.5

Who became the Nawab of Bengal after the death of Alivardi Khan in the year 1756?

(A) Wazir Ali
(B) Saadat Ali Khan
(C) Siraj-Ud-Daulah
(D) Shuja-Ud-Daulah

Q.6

Who among the following was the first woman President of the Indian National Congress?

(A) Sarojini Naidu
(B) Aruna Asaf Ali
(C) Sucheta Kriplani
(D) Annie Besant

Q.7

What is a 'double fault' in tennis?

(A) A serve that hits the net and goes into the correct service box
(B) A serve that goes out of bounds
(C) A fault called by the linesman
(D) Serving two faults in a row

Q.8

Birju Maharaj is related to which of the following classical dance forms?

(A) Garba
(B) Kathak
(C) Kathakali
(D) Bharatanatyam

Q.9

Kharavela was an important ruler of which dynasty?

(A) Vardhan Dynasty
(B) Chedi Dynasty
(C) Sunga Dynasty
(D) Gupta Dynasty

Q.10

Which of the following statements is/are true for the Religious Festivals in India? 1.Losar festival is majorly celebrated by Buddhist community of India. 2.Navroz festival is majorly celebrated by the Parsi community of India. 3.Saga Dawa is majorly celebrated by the Muslim community of India.

(A) Only 1
(B) Both 2 and 3
(C) Both 1 and 2
(D) Only 2

Q.11
Which of the following iron and steel plants is located in Jharkhand?

(A) Salem
(B) Durgapur
(C) Vijay Nagar
(D) Bokaro

Q.12
Which of the following peaks is NOT located in India?

(A) Nanda Devi
(B) Guru Shikar
(C) Mahendragiri
(D) Dhaulagiri

Q.13
With reference to medieval history, consider the following statements. 1. The founder of the Chola dynasty called the 'Imperial Cholas' was Vijayalaya (9th century AD). 2. One of the important dynasties which ruled in northern India was Gurjara-Pratiharas. 3. Palas ruled in western India and Rashtrakutas ruled in southern India in the early years (750 -1000). Which of the given statement(s) is/are correct?

(A) 1 only
(B) 1 and 2 only
(C) 2 only
(D) 1 and 3 only

Q.14
Which of the following is an informal group consisting of 4 to 10 individuals who seek loans against mutual guarantee?

(A) Self-help group
(B) Rural cooperatives
(C) Joint liability group
(D) Grameen model bank

Q.15
Who has been appointed as the new Chief of the Special Protection Group (SPG) by the centre in November 2023?

(A) Rakesh Asthana
(B) Ashok Kumar Mishra
(C) Anil Kumar Dhasmana
(D) Alok Sharma

Q.16
Who among the following elect(s) the presiding officer of the Vidhan Sabha in India?

(A) Governor
(B) Members of Vidhan Parishad
(C) President
(D) Members of Vidhan Sabha

Q.17
The major carbohydrate of milk is

(A) starch
(B) lactose
(C) Fructose
(D) maltose

Q.18
The national animal of Nepal is:

(A) tiger
(B) snow leopard
(C) camel
(D) cow

Q.19
Which of these state governments has implemented PESA act in November 2022?

(A) Madhya Pradesh
(B) Uttar Pradesh
(C) Rajasthan
(D) Punjab

Q.20
Which of the following Census recorded negative population growth in India?

(A) 1971
(B) 1991
(C) 2011
(D) 1921

Q.21
Which state in India hosted the 77th edition of the Santosh Trophy?

(A) Uttar Pradesh
(B) Madhya Pradesh
(C) Himanchal Pradesh
(D) Arunanchal Pradesh

Q.22
Who is the author of the historical fiction novel 'The Mauryan: The Legend of Ashoka'?

(A) Sanjeev Sanyal
(B) Amish Tripathi
(C) Chetan Bhagat
(D) Komal Bhanver

Q.23
Padma Bhushan awardee Mogubai Kurdikar was associated with which of the following music styles?

(A) Folk music
(B) Hindustani Classical music
(C) Theatre music
(D) Jazz music

Q.24
In which year did John Dalton present his atomic theory, the second part of which states that all atoms of a particular element are equal?

(A) 1811
(B) 1808
(C) 1825
(D) 1800

Q.25
The cold subarctic climate region possesses forest lands having conifers as main vegetation. These forest lands are known as:

(A) Taiga
(B) Tundra
(C) Mangrove
(D) Savanna

English Language and Comprehension 23 questions

Q.1
The following sentence has been divided into parts. One of them may contain an error. Select the part that contains the error from the given options. If you don't find any error, mark 'No error' as your answer. Whenever she is coming / here, she brings / many gifts for everyone.

(A) many gifts for everyone.
(B) No Error
(C) Whenever she is coming
(D) here, she brings

Q.2
Select the most appropriate option that can substitute the underlined segment in the given sentence. If there is no need to substitute it, select 'No substitution'. The pineapple dessert was "so delicious that we couldn't help eat it".

(A) so that delicious we couldn't help eating it
(B) so delicious that we couldn't helping eat it
(C) so delicious that we couldn't help eating it
(D) No substitution

Q.3

Select the most appropriate synonym of the underlined word in the following sentence. Don’t fight over such a “trivial” issue.

(A) Controversial
(B) Insignificant
(C) Complex
(D) Tricky

Q.4

Rectify the sentence by selecting the correct spelling from the options. After the storm passed, the sun emarged from behind the clouds.

(A) emerged
(B) amarged
(C) amergad
(D) emergad

Q.5

Select the INCORRECTLY spelt word.

(A) Rendezvous
(B) Lieutanant
(C) Bureaucracy
(D) Augment

Q.6

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. My sister performed / brilliant / in the competition / yesterday.

(A) in the competition
(B) my sister performed
(C) yesterday
(D) brilliant

Q.8

Select the most appropriate option to fill in the blank. My brother plays the guitar _____ than I do.

(A) better
(B) well
(C) good
(D) best

Q.9

Select the most appropriate option to fill in the blank. He is an intelligent person who always thinks before he _____.

(A) acted
(B) acting
(C) act
(D) acts

Q.10

The following sentence has been divided into parts. One of them may contain an error. Select the part that contains the error from the given options. If you don’t find any error, mark ‘No error’ as your answer. Ravi said that / his red jacket is / the cheaper than all other jackets.

(A) No error
(B) his red jacket is
(C) the cheaper than all other jackets.
(D) Ravi said that

Q.11

Select the most appropriate ANTONYM for the underlined word. They want to “bifurcate” the states.

(A) Support
(B) Divide
(C) Examine
(D) Combine

Q.12

Select the most appropriate meaning of the given idiom. Straight from the horse's mouth

(A) Rumours begin at home
(B) Directly from the person involved
(C) Giving no expression at all
(D) Directly eating from the provider

Q.13

Select the most appropriate option to fill in the blank. The assignment was _____ early in the morning.

(A) dew
(B) do
(C) due
(D) doe

Q.14

Select the most appropriate option to fill in the blank. The author's use of vivid imagery _____ the reader's senses, making the story come alive.

(A) heightens
(B) confuses
(C) actions
(D) dulls

Q.16

Select the appropriate option that can substitute the underlined word in the following sentence to make it grammatically correct. Smart watches are “the last” gadgets used by younger generations.

(A) later
(B) latest
(C) the latest
(D) the later

Q.17

Select the most appropriate ANTONYM of the given word: Grapple

(A) Clasp
(B) Release
(C) Grumble
(D) Wrestle

Q.18

Select the most appropriate option to fill in the blank. The way Sonam performs live on stage can be called the _____ one.

(A) more perfect
(B) most perfect
(C) perfectable
(D) mostly perfect

Q.19

Select the most appropriate synonym of the underlined word in the given sentence. He is a complete “novice” when it comes to driving a car.

(A) Zealous
(B) Swindler
(C) Expert
(D) Amateur

Q.20

Select the meaningful and correctly spelt word from the given options to fill in the blank. The oceans cover more than 70% of the Earth's surface, making them the largest _____ on the planet.

(A) habitude
(B) hybrid
(C) habit
(D) habitat

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank: In the realm of literature, words are the (1) _____ of creation, and authors are the (2) _____ of meaning. They skillfully weave stories with (3) _____ narratives that can (4) _____ the depths of human emotion. A well-crafted novel can (5) _____ readers into its world, making them forget the passage of time.

Select the most appropriate option to fill in blank number 1.

(A) tools
(B) mundane
(C) isolate
(D) convey

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank: In the realm of literature, words are the (1) _____ of creation, and authors are the (2) _____ of meaning. They skillfully weave stories with (3) _____ narratives that can (4) _____ the depths of human emotion. A well-crafted novel can (5) _____ readers into its world, making them forget the passage of time.

Select the most appropriate option to fill in blank number 2.

(A) keepers
(B) coward
(C) vibrators
(D) direction

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank: In the realm of literature, words are the (1) _____ of creation, and authors are the (2) _____ of meaning. They skillfully weave stories with (3) _____ narratives that can (4) _____ the depths of human emotion. A well-crafted novel can (5) _____ readers into its world, making them forget the passage of time.

Select the most appropriate option to fill in blank number 3.

(A) defrost
(B) mundane
(C) vivid
(D) convey

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank: In the realm of literature, words are the (1) _____ of creation, and authors are the (2) _____ of meaning. They skillfully weave stories with (3) _____ narratives that can (4) _____ the depths of human emotion. A well-crafted novel can (5) _____ readers into its world, making them forget the passage of time.

Select the most appropriate option to fill in blank number 4.

(A) teacher
(B) banker
(C) sailor
(D) pilot

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank: In the realm of literature, words are the (1) _____ of creation, and authors are the (2) _____ of meaning. They skillfully weave stories with (3) _____ narratives that can (4) _____ the depths of human emotion. A well-crafted novel can (5) _____ readers into its world, making them forget the passage of time.

Select the most appropriate option to fill in blank number 5.

(A) isolate
(B) immerse
(C) repel
(D) install

Answer key

Numerical and Mathematical Ability

Q.2 **(D)**	Q.3 **(A)**	Q.4 **(A)**	Q.5 **(D)**	Q.7 **(D)**	Q.8 **(A)**
Q.10 **(D)**	Q.11 **(C)**	Q.13 **(B)**	Q.15 **(D)**	Q.18 **(A)**	Q.19 **(D)**
Q.20 **(D)**					

Reasoning Ability and Problem Solving

Q.1 **(D)**	Q.4 **(D)**	Q.5 **(A)**	Q.6 **(C)**	Q.7 **(B)**	Q.9 **(A)**
Q.10 **(C)**	Q.11 **(A)**	Q.12 **(D)**	Q.14 **(D)**	Q.15 **(C)**	Q.16 **(B)**
Q.18 **(A)**	Q.19 **(A)**				

General Awareness

Q.1 **(D)**	Q.2 **(B)**	Q.3 **(C)**	Q.4 **(A)**	Q.5 **(C)**	Q.6 **(D)**
Q.7 **(D)**	Q.8 **(B)**	Q.9 **(B)**	Q.10 **(C)**	Q.11 **(D)**	Q.12 **(D)**
Q.13 **(B)**	Q.14 **(C)**	Q.15 **(D)**	Q.16 **(D)**	Q.17 **(B)**	Q.18 **(D)**
Q.19 **(A)**	Q.20 **(D)**	Q.21 **(D)**	Q.22 **(D)**	Q.23 **(B)**	Q.24 **(B)**
Q.25 **(A)**					

English Language and Comprehension

Q.1 **(C)**	Q.2 **(C)**	Q.3 **(B)**	Q.4 **(A)**	Q.5 **(B)**	Q.6 **(D)**
Q.8 **(A)**	Q.9 **(D)**	Q.10 **(C)**	Q.11 **(D)**	Q.12 **(B)**	Q.13 **(C)**
Q.14 **(A)**	Q.16 **(C)**	Q.17 **(B)**	Q.18 **(B)**	Q.19 **(D)**	Q.20 **(D)**
Q.21 **(A)**	Q.22 **(A)**	Q.23 **(C)**	Q.24 **(D)**	Q.25 **(B)**	

Explanations

Numerical and Mathematical Ability

Q.2 · Answer (D) 16.2

Total = 12 × 16.5 + 18 × 16 = 198 + 288 = 486 for 30 students: 486/30 = 16.2.

Q.3 · Answer (A) 40 min

The distance is 10 × 2 = 20 km. At 15 km/h it takes 20/15 = 4/3 hours = 80 minutes, 40 minutes less than 2 hours.

Q.4 · Answer (A) 61 kg

There are 64 females and 96 males: (64 × 58 + 96 × 63)/160 = (3,712 + 6,048)/160 = 9,760/160 = 61 kg.

Q.5 · Answer (D) 600

SI = 2,500 × 4 × 6/100 = ₹600.

Q.7 · Answer (D) 32%

Each article costs ₹15/15 = ₹1 and sells for ₹1.32, a profit of ₹0.32 on ₹1: 32%.

Q.8 · Answer (A) 41100

SI = 30,000 × (8 × 4 + 10 × 3 + 15 × 5)/100 = 30,000 × 137/100 = ₹41,100.

Q.10 · Answer (D) 36 years

Their ages total 2 × 48 = 96 years, so the son is 3/8 × 96 = 36.

Q.11 · Answer (C) 3.24 % decrease

1.18 × 0.82 = 0.9676 of the number is left: a fall of 3.24% ($x^2/100 = 18^2/100$).

Q.13 · Answer (B) 60 km/h

Distance = 65 × 4 + 52 × 2.5 = 260 + 130 = 390 km in 6.5 hours: 390/6.5 = 60 km/h.

Q.15 · Answer (D) 4.2 cm

The largest cone stands on one face, its base the circle inscribed in that face: radius 8.4/2 = 4.2 cm (height 8.4 cm).

Q.18 · Answer (A) ₹4,284

₹9,000 × 0.7 × 0.8 × 0.85 = ₹4,284.

Q.19 · Answer (D) 59

The average of consecutive numbers is the middle one, so the seven are 53 to 59; the greatest is 59.

Q.20 · Answer (D) 300

P = 240 × 100/(20 × 4) = ₹300.

Reasoning Ability and Problem Solving

Q.1 · Answer (D) 2 and 6

Interchanging 2 and 6: (12 ÷ 3) × 18 − 6 × 4 + 42 ÷ 2 = 72 − 24 + 21 = 69.

Q.4 · Answer (D) Neither conclusion (I) nor (II) follows.

Nothing links reptiles to snakes beyond the lizards, which are not snakes, so I does not follow; and all lizards being reptiles does not make all reptiles lizards, so II does not follow either.

Q.5 · Answer (A) mq

'friend' is the only word in both sentences and 'mq' the only code in both, so friend = mq.

Q.6 · Answer (C) 119

Each number is multiplied by 8 and 1 subtracted: 13 × 8 − 1 = 103 and 9 × 8 − 1 = 71. Likewise 15 × 8 − 1 = 119.

Q.7 · Answer (B) LSOV
Each letter moves 6 places forward: J → P, Q → W, M → S, T → Z, and HOKR → NUQX the same way. So F → L, M → S, I → O, P → V: LSOV.

Q.9 · Answer (A) Second to the left
With B second to C's left, D on C's other side (C and F between D and B), A third to F's right and E on G's left, the order going round to each person's left is C, F, B, G, E, A, D. E is second to the left of B.

Q.10 · Answer (C) MQT
Each letter moves 3 places forward each time: A, D, G, J, M; E, H, K, N, Q; H, K, N, Q, T. So MQT.

Q.11 · Answer (A) 3
M, O and T are in both words and 1, 4 and 5 in both codes. H, left in MOTH, is 2, and E, left in TOME, is 3.

Q.12 · Answer (D) 127
Each term is twice the one before plus 1: 63 × 2 + 1 = 127.

Q.14 · Answer (D) 87
The code is the sum of the letters' places in the alphabet: ROASTER = 18 + 15 + 1 + 19 + 20 + 5 + 18 = 96 and GUM = 7 + 21 + 13 = 41. ARTIST = 1 + 18 + 20 + 9 + 19 + 20 = 87.

Q.15 · Answer (C) 25
After interchanging: 29 ÷ 841 × 87 − 37 + 59 = 3 − 37 + 59 = 25 (841 = 29^2, so 29 ÷ 841 × 87 = 87/29 = 3).

Q.16 · Answer (B) 16, 128
The second number is 8 times the first: 11 × 8 = 88 and 14 × 8 = 112. Likewise 16 × 8 = 128.

Q.18 · Answer (A) Neither conclusion I nor II follows
Soaps and conditioners are both kept apart from lotions, which says nothing about soaps and conditioners, so I does not follow. As no lotion is a conditioner, no conditioner is a lotion, so II is false.

Q.19 · Answer (A) P
O is second to Q's left and N on Q's right; with P not beside Q and M second to L's right, the order going round to each person's left is Q, M, O, L, P, N. P sits between L and N.

General Awareness

Q.1 · Answer (D) Vitamin B12
Vitamin B12 (cobalamin) has a cobalt atom at the centre of its molecule.

Q.2 · Answer (B) Rao Jodha
Rao Jodha, the Rathore ruler who founded Jodhpur, began building Mehrangarh Fort in 1459.

Q.3 · Answer (C) V Satyanarayana Sarma
Vedantam Satyanarayana Sarma, a Kuchipudi dancer famous for female roles such as Satyabhama, received the Padma Shri in 1970.

Q.4 · Answer (A) 2002
The 86th Amendment Act, 2002, added Article 51A(k): parents and guardians are to provide opportunities for education to their children aged 6 to 14.

Q.5 · Answer (C) Siraj-Ud-Daulah
Siraj-ud-Daulah, Alivardi Khan's grandson, succeeded him in 1756; he lost Bengal to the East India Company at Plassey in 1757.

Q.6 · Answer (D) Annie Besant
Annie Besant presided over the Calcutta session of 1917; Sarojini Naidu, in 1925, was the first Indian woman President.

Q.7 · Answer (D) Serving two faults in a row
A double fault is failing with both serves on a point, which loses the server the point.

Q.8 · Answer (B) Kathak
Pandit Birju Maharaj, of the Lucknow gharana, was a leading exponent of Kathak.

Q.9 · Answer (B) Chedi Dynasty
Kharavela, the Kalinga king known from the Hathigumpha inscription, belonged to the Chedi (Mahameghavahana) dynasty.

Q.10 · Answer (C) Both 1 and 2
Losar, the Tibetan new year, is a Buddhist festival and Navroz the Parsi new year. Saga Dawa is also Buddhist, not Muslim, so 3 is false.

Q.11 · Answer (D) Bokaro
Bokaro Steel Plant is in Jharkhand; Durgapur is in West Bengal, Salem in Tamil Nadu and Vijayanagar in Karnataka.

Q.12 · Answer (D) Dhaulagiri
Dhaulagiri is in Nepal. Nanda Devi is in Uttarakhand, Guru Shikhar in Rajasthan and Mahendragiri in Odisha.

Q.13 · Answer (B) 1 and 2 only
Vijayalaya founded the Imperial Chola line in the 9th century, and the Gurjara-Pratiharas ruled in northern India. The Palas ruled in eastern India (Bengal and Bihar), not western, so 3 is wrong.

Q.14 · Answer (C) Joint liability group
NABARD describes a joint liability group as an informal group of 4 to 10 people who borrow from a bank against a mutual guarantee.

Q.15 · Answer (D) Alok Sharma
Alok Sharma, a 1991-batch IPS officer already serving in the SPG, was named its Director in November 2023, after the death of Arun Kumar Sinha.

Q.16 · Answer (D) Members of Vidhan Sabha
Under Article 178 the Legislative Assembly chooses its Speaker from among its own members.

Q.17 · Answer (B) lactose
Lactose, a disaccharide of glucose and galactose, is the sugar of milk.

Q.18 · Answer (D) cow
The cow is Nepal's national animal.

Q.19 · Answer (A) Madhya Pradesh
Madhya Pradesh brought its PESA rules into force on 15 November 2022, Janjatiya Gaurav Divas.

Q.20 · Answer (D) 1921
1921 is the only census to record a fall in India's population, after the influenza epidemic of 1918 and famines; it is called the year of the great divide.

Q.21 · Answer (D) Arunanchal Pradesh
The final round of the 77th Santosh Trophy (2023-24) was played at Yupia in Arunachal Pradesh; Services won it.

Q.22 · Answer (D) Komal Bhanver
'The Mauryan: The Legend of Ashoka' (2017) is a historical novel by Komal Bhanver.

Q.23 · Answer (B) Hindustani Classical music
Mogubai Kurdikar was a Hindustani classical singer of the Jaipur-Atrauli gharana.

Q.24 · Answer (B) 1808
Dalton set out his atomic theory in 'A New System of Chemical Philosophy' in 1808: matter is made of atoms, and the atoms of one element are identical in mass and properties.

Q.25 · Answer (A) Taiga
The taiga, or boreal forest, is the belt of coniferous forest in the cold subarctic climate; the tundra north of it is largely treeless.

English Language and Comprehension

Q.1 · Answer (C) Whenever she is coming
A repeated, habitual action takes the simple present: 'Whenever she comes here'.

Q.2 · Answer (C) so delicious that we couldn’t help eating it
'Can't help' is followed by a gerund: 'couldn't help eating it'.

Q.3 · Answer (B) Insignificant
'Trivial' means of little importance: 'insignificant'.

Q.4 · Answer (A) emerged
The correct spelling is 'emerged'.

Q.5 · Answer (B) Lieutanant
The correct spelling is 'Lieutenant'.

Q.6 · Answer (D) brilliant
A verb is modified by an adverb: 'My sister performed brilliantly'.

Q.8 · Answer (A) better
'Than' calls for the comparative, and the comparative of the adverb 'well' is 'better'.

Q.9 · Answer (D) acts
The habitual present 'always thinks' is matched by the simple present with 'he': 'before he acts'.

Q.10 · Answer (C) the cheaper than all other jackets.
A comparative with 'than' takes no 'the': 'cheaper than all other jackets'.

Q.11 · Answer (D) Combine
To 'bifurcate' is to divide into two; its opposite is 'combine'.

Q.12 · Answer (B) Directly from the person involved
Information 'straight from the horse's mouth' comes directly from the person involved, the most reliable source.

Q.13 · Answer (C) due
An assignment that is 'due' at a time must be handed in then; 'dew' (moisture), 'do' and 'doe' (a female deer) make no sense here.

Q.14 · Answer (A) heightens
Imagery that makes a story come alive 'heightens', or sharpens, the reader's senses.

Q.16 · Answer (C) the latest
'The latest' means the newest; 'the last' means the final one: 'Smart watches are the latest gadgets'.

Q.17 · Answer (B) Release
To 'grapple' is to seize and hold, or struggle with; its opposite is 'release'.

Q.18 · Answer (B) most perfect
SSC's key gives 'most perfect': 'the … one', comparing her way of performing with all others, calls for the superlative, and 'more perfect' is a comparative while 'perfectable' and 'mostly perfect' do not fit. Strictly, 'perfect' is an absolute adjective that takes no degrees, though 'most perfect' is common in use.

Q.19 · Answer (D) Amateur
A 'novice' is a beginner without experience: an 'amateur'.

Q.20 · Answer (D) habitat
A 'habitat' is the natural home of living things; 'habitude' means a habit, and 'hybrid' a cross.

Q.21 · Answer (A) tools
Words are the 'tools', the instruments, with which literature is created.

Q.22 · Answer (A) keepers
Authors are the 'keepers' of meaning, those who hold and shape it.

Q.23 · Answer (C) vivid
Before 'narratives' the blank needs an adjective: 'vivid', lively and clear. 'Mundane' (dull) would contradict the praise, and 'defrost' and 'convey' are verbs.

Q.24 · Answer (D) pilot
After 'can' the blank needs a verb, and of the options only 'pilot' is one: to steer or navigate through the depths of emotion.

Q.25 · Answer (B) immerse
A novel can 'immerse', or absorb, readers in its world, so that they forget the passage of time.

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